

ALLA KRISHNA SAI REDDY

AI/ML Engineer | Generative AI | Full Stack Developer (Python, TypeScript)

Vijayawada, Andhra Pradesh 521212 — +91 9542221188 — krishnasai2004reddy@gmail.com

Portfolio — LinkedIn — GitHub — huggingface.co

PROFESSIONAL SUMMARY

AI/ML and full stack engineer who builds LLM applications end to end: LangGraph multi-agent workflows, Retrieval-Augmented Generation (RAG), OCR-based document extraction, and REST APIs in Python (FastAPI, Flask) and Node.js/TypeScript (Express, Next.js). Deployed applications on Vercel, Render, and Hugging Face Spaces. Author of one published paper on deep learning for medical image classification.

EDUCATION

Bachelor of Technology in Computer Science and Engineering (AI & ML)

2021 — 2025

Lovely Professional University, Jalandhar, Punjab

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, HTML, CSS, Data Structures & Algorithms

Full-stack development & APIs: React 18, Next.js 14, Tailwind CSS, FastAPI, Flask, Node.js, Express.js, RESTful APIs

Generative AI & LLMs: Retrieval-Augmented Generation (RAG), LangGraph, LLM APIs, Mistral OCR

Machine Learning: PyTorch, TensorFlow, OpenCV, NumPy, Deep Learning, CNNs, Transfer Learning, Computer Vision

Databases & Vector Search: PostgreSQL, Supabase, pgvector, FAISS, Vector Embeddings, MySQL, MongoDB, SQLite

Cloud & Tools: Docker, GCP, Hugging Face Spaces, Cloudflare Workers, Git, GitHub, Postman, Manual Software Testing

EXPERIENCE

Freelance Full Stack Developer — Self-Employed, Vijayawada, India

Aug, 2025 – Present

Manohar Organic Spices (client website) | Astro, JavaScript, Tailwind CSS, Cloudflare Workers

Live: manoharorganicspices.com

- Built and deployed a responsive multi-page static site with reusable components, shared layouts, and a centralized data layer for product management.
- Implemented technical SEO (JSON-LD structured data, XML sitemap, meta tags) and validated the schema with Google Rich Results Test.
- Deployed on Cloudflare Workers with a custom domain, environment variables, and image optimization.

PROJECTS

Klerk AI – Autonomous Administrative Automation Suite - Node.js, Express.js, Next.js 14, TypeScript, PostgreSQL, GCP

Live Demo: vercel.com | GitHub: github.com

- Built an AI assistant that extracts invoices, receipts, and quotes from WhatsApp and Gmail messages and stores them as structured records, automating back-office paperwork for French tradespeople.
- Implemented document understanding with Mistral OCR and a hybrid RAG router; added validation rules and duplicate detection to block bad and repeated records.
- Developed REST APIs in Express.js with asynchronous workers and PostgreSQL storage, deployed to production on Vercel and Render.

JobJutsu AI – AI-Powered Career Copilot - Python, LangGraph, Streamlit, Hugging Face Spaces, Gemini (LLM)

Live Demo: huggingface.co/spaces | GitHub: github.com

- Built a LangGraph multi-agent application that parses resumes, matches them to job descriptions, identifies skill gaps, and generates role-specific interview questions with Gemini.
- Designed modular agent workflows with orchestration and conditional routing across resume analysis, job matching, gap detection, and question generation.
- Containerized with Docker and deployed on Hugging Face Spaces as a live public web application.

AI-Powered Detection of Diabetic Retinopathy - Python, PyTorch, TensorFlow, Flask, CV, NumPy, SQLite

GitHub: github.com | Publication: publication

- Built a deep learning image classifier for retinal images, reaching 82% validation accuracy.
- Served the model through a Flask web application and API layer with SQLite storage.
- Ran functional, regression, and smoke tests on the API layer and fixed defects; published results in IRJET, Vol. 12, Issue 05 (May 2025).